Topic 39 — Acid Rain (अम्ल वर्षा)

Covers syllabus (click to expand)

Acid Rain | Causes | Effects | Prevention

Sources baked in: NCERT; CPCB; CLRTAP/Gothenburg; Taj Trapezium; UPPCS PYQs **Weight:** ★★ Medium–High — pH traps; CO_2 vs SO_2/NOx; Taj gypsum; Montreal ≠ acid rain **Last verified:** August 2026

Current Affairs (this topic)

- Twin of Acid Rain core — chemistry fixed; rare CA only.
- Acid rain ≠ photochemical smog ≠ ozone depletion (ओजोन क्षरण).

39.1 Acid Rain

Identity: pH < 5.6 | H_2SO_4 + HNO_3 | Wet + dry | Secondary pollutant

- **Acid rain** means precipitation or atmospheric deposition with **pH below 5.6**, mainly from **sulphuric acid (H_2SO_4)** and **nitric acid (HNO_3)** (नाइट्रिक अम्ल).
- **Normal unpolluted rain** is already slightly acidic (~**pH 5.6**) because **CO_2** forms weak **carbonic acid (H_2CO_3)** — it is **not** pH 7.
- Anthropogenic episodes can fall to about **pH 4.0–4.5** (or lower in extreme cases).
- Includes **wet deposition** (rain, snow, fog) and **dry deposition** (acidic gases/particles settling without rain).
- Acids form in the air, so acid rain is a **secondary pollution** problem.
- On the pH scale: **acidic < 7**, **neutral = 7**, **basic > 7**. Each step down is **10×** more H^+.

PYQ — UPPCS Prelims 2025, Q103

1. Acidic solutions have pH more than 7.
2. Basic solutions have pH less than 7.

Options: A. Only 2 B. Neither 1 nor 2 C. Both D. Only 1

Show answer

Ans: B — Acidic **< 7**; basic **> 7**.

PYQ — UPPCS Prelims 2024, Q108

(A) CO_2, SO_2, and NO_2 dissolve in rainwater and produce acid rain. (R) Air is polluted due to high CO_2, SO_2, and oxides of nitrogen.

Options: A. Both true; R explains A B. A false, R true C. Both true; R not explanation D. A true, R false

Show answer

Ans: C — All three dissolve (A treated true). R is true about air pollution but does **not** explain why **CO_2 gives only weak acid** while **SO_2/NOx drive strong acid rain**.

pH	Meaning
< 5.6	Acid rain threshold
~5.6	Normal rain (CO_2)
7	Neutral (not natural rain)
> 7	Basic

> **Logic:** Normal rain ≠ neutral. CO_2 alone ≠ severe acid rain.

39.2 Causes

Identity: SO_2 → H_2SO_4 | NOx → HNO_3 | Coal TPPs + vehicles | CO_2 weak only

- **SO_2** from burning sulphur-rich **coal/oil** in thermal plants, refineries, smelters, brick kilns, and generators forms **H_2SO_4**.
- **NOx** from vehicles, power plants, and high-temperature combustion forms **HNO_3**.
- Natural sources (volcanoes for SO_2; lightning for NOx) exist but are minor versus anthropogenic loads in industrial regions.
- **CO_2** only sets the weak baseline (~5.6); it is **not** the main cause of damaging acid rain.
- **SO_2 is heavier than air** and can linger near ground-level sources.
- India/UP hotspots: Singrauli thermal belt, Mathura (मथुरा) refinery–Agra corridor, industrial NCR/Kanpur (कानपुर) airsheds.

PYQ — UPPCS Prelims 2025, Q43 (pattern)

Gases lighter than air among listed options — only **helium** fits; **SO_2 and HCl are heavier**.

> **Logic:** SO_2 sinks relative (सापेक्ष) to air — density trap.

39.3 Effects

Identity: Aquatic kill | Soil Al^{3+} | Conifers | Taj gypsum | Lichens

- Freshwater life fails when pH falls below about **5.0** — fish eggs and many aquatic species die; toxic **Al^{3+}** is mobilised.
- Soils lose **Ca^{2+}/Mg^{2+}** nutrients and gain toxic aluminium, damaging roots.
- **Conifers** (pine, spruce) are especially sensitive to acid fog and soil change.
- Marble/limestone monuments: **$CaCO_3$ + acid → gypsum ($CaSO_4 \cdot 2H_2O$) crust** — classic **Taj Mahal** (ताज महल) yellowing/damage.
- Taj damage is linked to **SO_2** from Mathura refinery and nearby industries, plus particulates.
- Human health: SO_2/NOx and acid aerosols irritate lungs.
- **Lichens** die in high-SO_2 air and act as pollution bio-indicators (Topic 38 crossover).

Receptor	Effect
Lakes/rivers	Fish kills below ~pH 5
Soil	Nutrient leach + Al^{3+}
Forests	Conifer damage
Marble	Gypsum crust (Taj)
Lichens	Die in high SO_2

> **Logic:** Taj fact = **SO_2/acid**, not "only CO_2 yellowing."

39.4 Prevention

Identity: FGD | Catalytic converters | BS-VI | Gothenburg | Taj Trapezium

- **Flue Gas Desulphurisation (FGD)** scrubbers use lime/limestone to cut SO_2 from power-plant stacks.
- Vehicle **catalytic converters** and cleaner fuels cut NOx; **BS-VI** fuel has very low sulphur (~**10 ppm**).
- Switch from coal/diesel to **CNG/LPG/renewables** reduces acidifying emissions.
- CPCB (केंद्रीय प्रदूषण नियंत्रण बोर्ड) emission standards and consents limit SO_2/NOx from industry.
- International: **Gothenburg Protocol (1999)** (गोथेनबर्ग प्रोटोकॉल) under **CLRTAP** targets acidifying and related air pollutants — **not** the Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल) (ozone).
- **Taj Trapezium Zone (~10,400 km²)** — Supreme Court-driven controls (1996 onward) forced gas switch/relocation of polluting units around Agra.

Tool	Cuts
FGD	SO_2
Catalytic converter / BS-VI	NOx + fuel S
Fuel switch	Both
Taj Trapezium	Local heritage protection

Tool	Cuts
Gothenburg/CLRTAP	Transboundary acidification

> **Logic:** Montreal = ozone. Acid rain treaty family = **CLRTAP / Gothenburg**.
